ApexUCL Tournament Management System

Deliverable 3: Use Case Diagram & Fully Dressed Use Cases
Course: Software Design & Analysis (SDA)

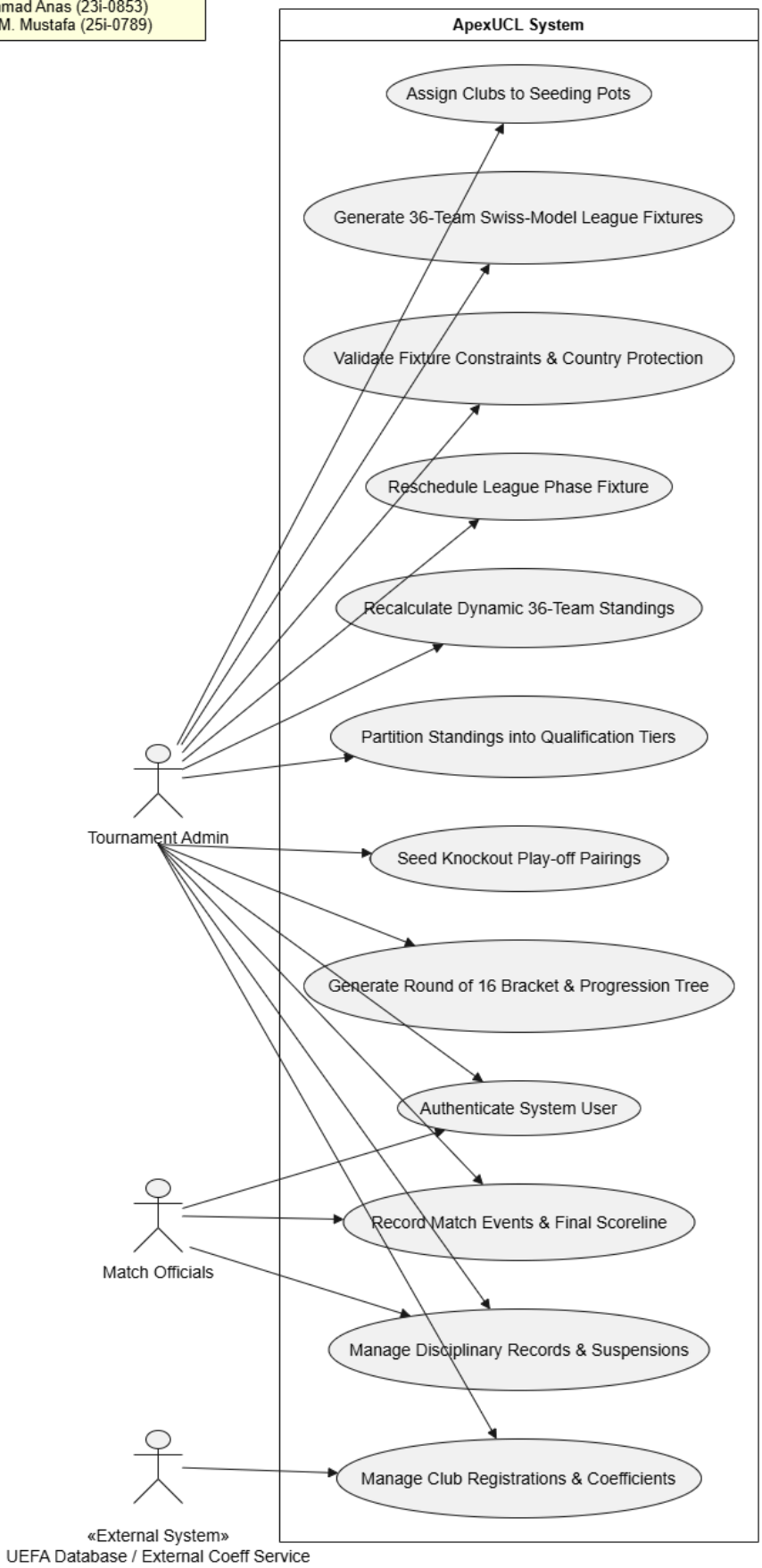
1. Team Contributions & Role Distribution

Team Member	Roll Number	Assigned Roles	D3 Specific Contributions
Muhammad Anas	23i-0853	Team Manager, Lead Analyst, UI/UX Designer	• System boundary scoping & actor hierarchy formulation • Granularity governance (<=2 Manage rules) • Cross-use-case verification & final consolidation
Aazan Muhammad Mustafa	25i-0789	Systems Analyst, QA Manager	• Modeled Use Case Diagram in Draw.io • Authored all 12 Fully Dressed Use Cases (UC-01 to UC-12) • Mapped 2-column Main Success Scenarios • Documented extensions & edge conditions

2. Part One: Use Case Diagram

The project Use Case Diagram was modeled using Draw.io adhering to UML 2.0 standards. It defines the ApexUCL System boundary, the actors involved, and all 12 core functional goals.

Team Members:
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3. Part Two: 12 Fully Dressed Use Cases

Use Case 01: Authenticate System User

Use Case Name:	Authenticate System User
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin, Match Officials
Stakeholders & Interests:	• Admin/Officials: Want secure, rapid access to their authorized dashboards. • UEFA Governance: Ensures data integrity and prevents unauthorized modification of league results.
Preconditions:	The user account exists in the system database with an assigned role (Admin or Official).
Postconditions:	The user is authenticated and the role-specific graphical interface is rendered.

Main Success Scenario

Actor Action	System Response
1. User launches the ApexUCL desktop application.	
	2. System renders the centralized authentication portal.
3. User enters their designated user ID and password credentials.	
	4. System validates credentials against the local database schema via secure hash verification.
	5. System grants role-based access and loads the respective administrative or matchday official dashboard.

Extensions:

• 4a. Invalid credentials entered: System displays error message and prompts retry.
• 4b. Account locked after 3 failed attempts: System notifies user to contact technical support.

Special Requirements:

Authentication request must be evaluated within 1.0 second. Passwords must not be logged in plain text.

Technology & Data Variations List:

Relational storage queried via JDBC (MySQL/Oracle/SQL Server); JavaFX/Swing UI.

Use Case 02: Manage Club Registrations & Coefficients (Manage Use Case #1)

Use Case Name:	Manage Club Registrations & Coefficients (Manage Use Case #1)
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Tournament Admin: Needs to configure the 36 participating clubs accurately. • Clubs: Need authentic 5-year UEFA coefficient values to secure legitimate seeding pot tier.
Preconditions:	Tournament Admin is authenticated and logged into the administrative dashboard.
Postconditions:	All 36 clubs are registered with valid country associations and 5-year coefficient scores in the database.

Main Success Scenario

Actor Action	System Response
1. Admin accesses Club Management module and clicks 'Register Clubs'.	
	2. System opens the club registration interface.
3. Admin enters club name, country association code, and clicks 'Fetch Coefficient'.	
	4. System dispatches request to external UEFA Database / External Coeff Service.
	5. External Service transmits official 5-year historical coefficient score.
6. Admin reviews the fetched data and confirms the entry.	
	7. System commits club record into relational database and updates registered count.

Extensions:

• 4a. External Coeff Service unreachable: System prompts Admin to enter coefficient manually.
• 7a. Duplicate club entry: System detects existing registration and prompts Admin to update or cancel.

Special Requirements:

System must restrict registration strictly to 36 qualified clubs.

Technology & Data Variations List:

Data fetched via REST API; stored locally in SQL via JDBC.

Use Case 03: Assign Clubs to Seeding Pots

Use Case Name:	Assign Clubs to Seeding Pots
Scope:	ApexUCL System
Level:	Subfunction / User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Tournament Admin: Wants automated, transparent sorting to avoid human pot-allocation errors. • Clubs: Demand seeding strictly governed by official UEFA coefficient criteria.
Preconditions:	Exactly 36 clubs are registered with valid coefficient rankings in the system database.
Postconditions:	Clubs are partitioned into four distinct pots of 9 clubs each (Pots 1-4) and locked in database.

Main Success Scenario

Actor Action	System Response
1. Admin initiates 'Generate Seeding Pots' function.	
	2. System queries all 36 clubs from the relational database.
	3. System sorts clubs in descending order based on UEFA coefficient values.
	4. System partitions sorted list into 4 pots (Pots 1-4) with 9 teams each.
	5. System displays the categorized 4-pot visual

	layout on GUI dashboard.
6. Admin reviews and clicks 'Lock Pot Allocations'.	
	7. System updates pot assignment fields in database and marks pot generation finalized.

Extensions:

• 3a. Coefficient tie at pot boundary (e.g. 9th vs 10th): System prompts Admin to apply manual UEFA tie-breaker rules or automated draw.
• 4a. Reigning Champion Exception: UCL titleholder is placed into Pot 1 Slot 1 regardless of coefficient.

Special Requirements:

Complete sorting and pot-partitioning pipeline must execute in under 500 milliseconds.

Technology & Data Variations List:

In-memory sorting via Java Collections API; rendered using JavaFX TableViews/Swing.

Use Case 04: Generate 36-Team Swiss-Model League Fixtures

Use Case Name:	Generate 36-Team Swiss-Model League Fixtures
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Tournament Admin: Requires a 100% mathematically valid 144-match tournament schedule. • Clubs: Require fair venue parity (4 Home, 4 Away) and balanced pot exposure (2 per pot). • Broadcasters: Need early, immutable, conflict-free match schedules.
Preconditions:	Clubs are successfully assigned and locked into the four 9-team seeding pots.
Postconditions:	Exactly 144 league phase fixtures are generated, verified, and saved with status SCHEDULED.

Main Success Scenario

Actor Action	System Response
1. Admin opens Fixture Management and clicks 'Generate Swiss-Model Fixtures'.	

	2. System initializes the constraint-satisfaction fixture generation engine.
	3. System pairs each club with 2 opponents from each of Pots 1, 2, 3, and 4 (8 opponents per club).
	4. System balances venues so each team plays 4 Home and 4 Away matches (1 Home, 1 Away per pot).
	5. System renders the draft 144-fixture tournament matrix on the desktop grid.
6. Admin reviews generated schedule and clicks 'Approve & Commit Schedule'.	
	7. System persists the 144 fixtures into the Fixtures table with status SCHEDULED.

Extensions:

• 3a. Algorithmic deadlock during pairing: Backtracking logic automatically resets and recalculates valid paths with alternate seed.

Special Requirements:

Engine must utilize backtracking constraint satisfaction capable of generating full schedule in under 5 seconds.

Technology & Data Variations List:

Core logic executed in Java; visualized in a desktop fixture matrix grid.

Use Case 05: Validate Fixture Constraints & Country Protection

Use Case Name:	Validate Fixture Constraints & Country Protection
Scope:	ApexUCL System
Level:	Subfunction
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• UEFA Governance: Demands zero country protection violations and strict adherence to foreign association limits.
Preconditions:	A draft 144-match league schedule has been generated by the system.
Postconditions:	The schedule is certified as mathematically and legally compliant with

	UEFA regulations.

Main Success Scenario

Actor Action	System Response
1. Admin selects 'Run Compliance & Validation Audit' on generated schedule.	
	2. System initiates automated rule-checking across all 144 fixtures.
	3. System verifies no team plays an opponent from their own domestic league (Country Protection).
	4. System verifies no team plays more than 2 opponents from any single foreign association.
	5. System verifies all 36 clubs have exact 4-Home and 4-Away allocation.
	6. System presents 'Validation Successful: 100% UEFA Compliant' certification report.

Extensions:

• 3a/4a. Constraint violation detected: System flags specific clashes in red on GUI and prompts schedule recalculation.

Special Requirements:

Complete 144-fixture rule verification must execute in <= 1.0 second.

Technology & Data Variations List:

Automated validation scripts execute assertions against in-memory graph models.

Use Case 06: Reschedule League Phase Fixture

Use Case Name:	Reschedule League Phase Fixture
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Admin/Clubs: Need operational flexibility to reschedule matches due to weather or stadium conflicts.

Preconditions:	The 144-match fixture list is finalized, saved, and in SCHEDULED status.
Postconditions:	Targeted fixture's date/time/venue is updated without violating rest constraints.

Main Success Scenario

Actor Action	System Response
1. Admin opens Fixture Matrix Grid and selects fixture to modify.	
2. Admin clicks 'Reschedule Match'.	
	3. System loads current match metadata and opens reschedule dialog.
4. Admin enters new date, kick-off timestamp, and venue details.	
	5. System evaluates new parameters against both clubs' calendars to ensure 48-hour recovery rule.
	6. System updates fixture record in database with status RESCHEDULED.
	7. System updates GUI grid and displays confirmation.

Extensions:

• 5a. Date clash or inadequate rest interval: System flags violation and prompts for alternative date.

Special Requirements:

Updates must use atomic database transactions (COMMIT/ROLLBACK) via JDBC to guarantee schedule consistency.

Technology & Data Variations List:

Executed via parameterized SQL UPDATE statements on Fixtures table.

Use Case 07: Record Match Events & Final Scoreline

Use Case Name:	Record Match Events & Final Scoreline
Scope:	ApexUCL System

Level:	User-Goal
Primary Actor:	Match Officials, Tournament Admin
Stakeholders & Interests:	• Match Officials: Need responsive match-logging console. • Fans/Media: Rely on immediate final scorelines for live standings updates.
Preconditions:	Selected fixture exists in SCHEDULED or IN_PROGRESS status.
Postconditions:	Match scoreline is stored persistently, status transitions to FINAL, and table updates are triggered.

Main Success Scenario

Actor Action	System Response
1. Match Official navigates to Live Matchday Console and selects designated fixture.	
	2. System opens match event logger interface.
3. Official enters final Home Goals and Away Goals, and logs key match events.	
4. Official sets completion state to 'Final' and clicks 'Submit Result'.	
	5. System displays verification summary modal.
6. Official confirms submission.	
	7. System writes scoreline and telemetry into MatchResults table and updates status to COMPLETED.
	8. System triggers automated observer event to recalculate dynamic standings.

Extensions:

• 4a. Missing or negative values: System highlights invalid inputs in red and prevents submission.

Special Requirements:

Must apply GoF Observer Pattern to broadcast match completion events to standings engine.

Technology & Data Variations List:

Persisted to MySQL/Oracle MatchResults table via JDBC prepared transactions.

Use Case 08: Recalculate Dynamic 36-Team Standings

Use Case Name:	Recalculate Dynamic 36-Team Standings
Scope:	ApexUCL System
Level:	Subfunction
Primary Actor:	System (triggered automatically) / Tournament Admin
Stakeholders & Interests:	• All Users: Depend on mathematically accurate single-table standing sorted by official UEFA tie-breakers.
Preconditions:	A new final match scoreline has been committed with status COMPLETED.
Postconditions:	Standings across all 36 teams are recalculated, sorted, and refreshed on dashboard.

Main Success Scenario

Actor Action	System Response
1. System receives match completion trigger (or Admin clicks 'Recalculate Table').	
	2. System updates Points (Win: 3, Draw: 1, Loss: 0), Goal Difference, Goals Scored, Away Goals, and Wins.
	3. System invokes GoF Strategy Pattern (RankingStrategy).
	4. Strategy evaluates multi-tier tie-breakers: Points -> GD -> GF -> Away Goals -> Wins -> Away Wins -> Coeff.
	5. System establishes distinct ranks from 1 to 36.
	6. System refreshes Desktop Governance Console table view with new rankings.

Extensions:

• 4a. Absolute tie across all metric tiers: System applies final tie-breaker (5-year UEFA Club Coefficient).

Special Requirements:

Must strictly follow GoF Strategy Design Pattern to decouple tie-breaker sorting algorithms.

Technology & Data Variations List:

Java Comparator<StandingEntry> implementing custom sorting logic; StandingsEntry table.

Use Case 09: Manage Disciplinary Records & Suspensions (Manage Use Case #2)

Use Case Name:	Manage Disciplinary Records & Suspensions (Manage Use Case #2)
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Match Officials, Tournament Admin
Stakeholders & Interests:	• Tournament Admin & Clubs: Need accurate tracking of card accumulations and active player suspension lists.
Preconditions:	A league match is active or concluded.
Postconditions:	Cards are logged in player registry, and suspension flags are applied for subsequent fixtures.

Main Success Scenario

Actor Action	System Response
1. Official navigates to Disciplinary Incident Panel for a match.	
2. Official selects player, team, minute, and card infraction type (Yellow/Red).	
	3. System commits disciplinary record to relational database.
	4. System evaluates player's cumulative card count across 8-match phase.
	5. System identifies when suspension threshold is met (e.g. 3 yellows or 1 direct red).
	6. System automatically flags player profile as SUSPENDED for subsequent matchday.
	7. System updates official public suspension bulletin on administrative console.

Extensions:

• 2a. Non-rostered player entry: System rejects input and prompts for corrected selection.

Special Requirements:

Must track cumulative statistics accurately across the 8-match league phase.

Technology & Data Variations List:

Relational tracking via DisciplinaryRecords table joined with Players and Clubs.

Use Case 10: Partition Standings into Qualification Tiers

Use Case Name:	Partition Standings into Qualification Tiers
Scope:	ApexUCL System
Level:	Subfunction
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Admin, Media, Fans: Need instant visual clarity regarding who qualifies directly, proceeds to play-offs, or is eliminated.
Preconditions:	All 144 league phase fixtures are completed, and final 36-team standings are locked.
Postconditions:	Standings are partitioned into three official UEFA qualification tiers with distinct color coding on GUI.

Main Success Scenario

Actor Action	System Response
1. Admin navigates to finalized 36-team tournament dashboard.	
	2. System evaluates ranks 1-8, assigns tag DIRECT_R16, and color-codes rows Green.
	3. System evaluates ranks 9-24, assigns tag KNOCKOUT_PLAYOFF, and color-codes rows Blue.
	4. System evaluates ranks 25-36, assigns tag ELIMINATED, and color-codes rows Red.
	5. System displays formatted tripartite dashboard and generates stage-completion metadata.

Extensions:

• 2a/3a/4a. UI Color rendering failure: System falls back to explicit text badges ([R16 Bye], [Play-off], [Eliminated]).

Special Requirements:

Boundary Value Analysis must ensure precision at 8th vs 9th rank and 24th vs 25th rank margins.

Technology & Data Variations List:

Dynamic row styling via JavaFX CSS pseudoclasses or Swing TableCellRenderer.

Use Case 11: Seed Knockout Play-off Pairings

Use Case Name:	Seed Knockout Play-off Pairings
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Tournament Admin & Clubs: Need accurate two-legged pairings between seeded (9-16) and unseeded (17-24) clubs.
Preconditions:	Standings table has finalized the tripartite partition (identifying clubs in Ranks 9 to 24).
Postconditions:	Exactly 8 two-legged knockout play-off fixtures (16 total matches) are generated and saved.

Main Success Scenario

Actor Action	System Response
1. Admin clicks 'Generate Knockout Play-off Pairings'.	
	2. System separates the 16 play-off clubs into Seeded (9-16) and Unseeded (17-24) pools.
	3. System pairs Seeded vs Unseeded clubs using automated bracket engine.
	4. System schedules two-legged fixtures, assigning Leg 2 home match to Seeded club.
	5. System displays draft 8 play-off ties on admin dashboard.
6. Admin confirms pairings.	
	7. System commits records into KnockoutPairings table.

Extensions:

• 3a. Same-Association clash policy: Pairing engine adjusts ties if UEFA configuration mandates domestic separation.

Special Requirements:

Must utilize GoF Factory Pattern to instantiate TwoLeggedKnockoutTie objects.

Technology & Data Variations List:

Stored in KnockoutPairings table with leg1_home, leg2_home, and aggregate_score fields.

Use Case 12: Generate Round of 16 Bracket & Progression Tree

Use Case Name:	Generate Round of 16 Bracket & Progression Tree
Scope:	ApexUCL System
Level:	User-Goal
Primary Actor:	Tournament Admin
Stakeholders & Interests:	• Tournament Admin: Needs an interactive, fixed-path tournament bracket leading directly to the Final. • Broadcasters/Fans: Need clear visualization of bracket paths.
Preconditions:	The 8 two-legged Knockout Play-off ties have concluded with aggregate winners determined.
Postconditions:	Round of 16 bracket is fully populated and rendered as an interactive progression tree.

Main Success Scenario

Actor Action	System Response
1. Admin clicks 'Initialize Round of 16 Progression Tree'.	
	2. System retrieves 8 Direct Qualifiers (Ranks 1-8).
	3. System retrieves 8 advancing aggregate winners from Knockout Play-offs.
	4. System maps teams into predefined Round of 16 slots using Factory Pattern (Top 8 host Leg 2).
	5. System renders interactive graphical tournament tree (R16 -> QF -> SF -> Final).

6. Admin reviews progression tree and locks bracket paths.

Extensions:

• 3a. Play-off tie ends in aggregate draw: System checks Extra Time and Penalty Shootout inputs to advance winner.

Special Requirements:

Must use Factory Design Pattern to dynamically instantiate bracket stages.

Technology & Data Variations List:

Rendered interactively using JavaFX tree components / Swing; persisted in KnockoutPairings.
